Hybrid Model Output

1)

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==================================================
MODEL DEFINITION AND TRAINING
==================================================
🔧 Total models to test: 5

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MODEL TRAINING AND EVALUATION
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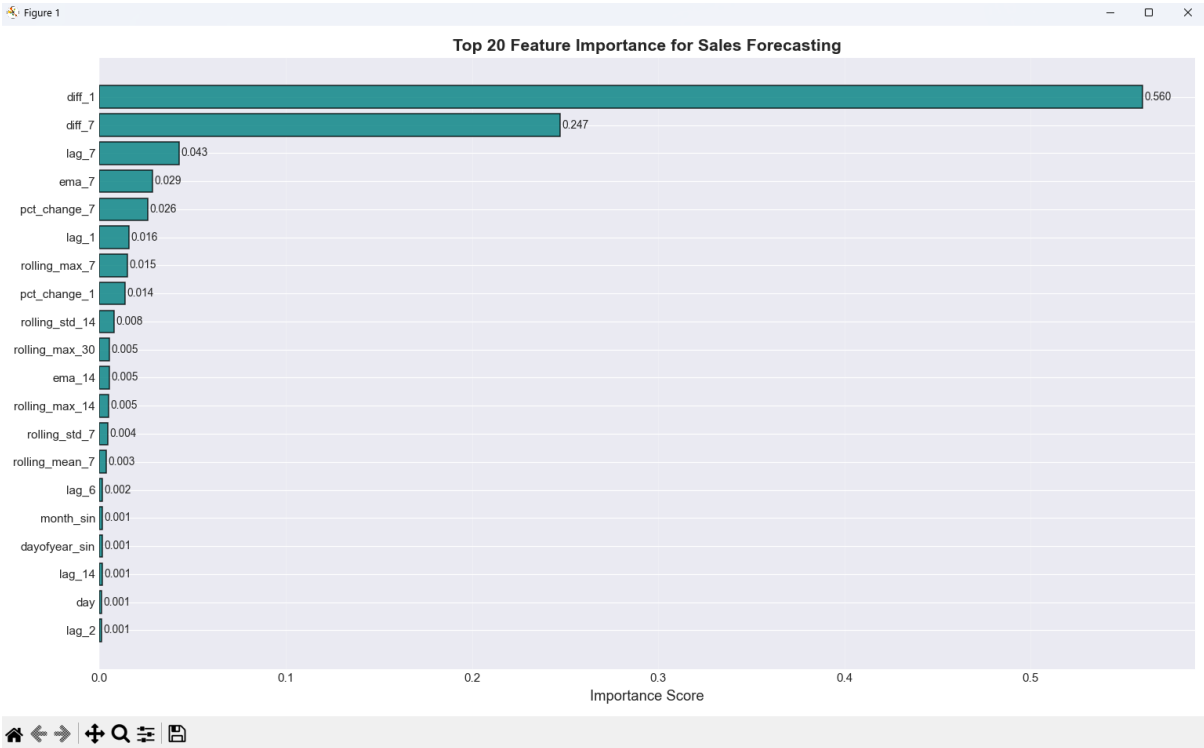
▶ Training Random Forest...
  ☑ MAE: $273.36 | RMSE: $456.69 | R²: 0.9691

▶ Training XGBoost...
  ☑ MAE: $286.09 | RMSE: $451.28 | R²: 0.9698

▶ Training LightGBM...
  ☑ MAE: $365.87 | RMSE: $682.14 | R²: 0.9311

▶ Training Gradient Boosting...
  ☑ MAE: $238.93 | RMSE: $480.10 | R²: 0.9658

▶ Training Ridge Regression...
  ☑ MAE: $16.43 | RMSE: $22.61 | R²: 0.9999
```



2)

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DETAILED FEATURE ANALYSIS
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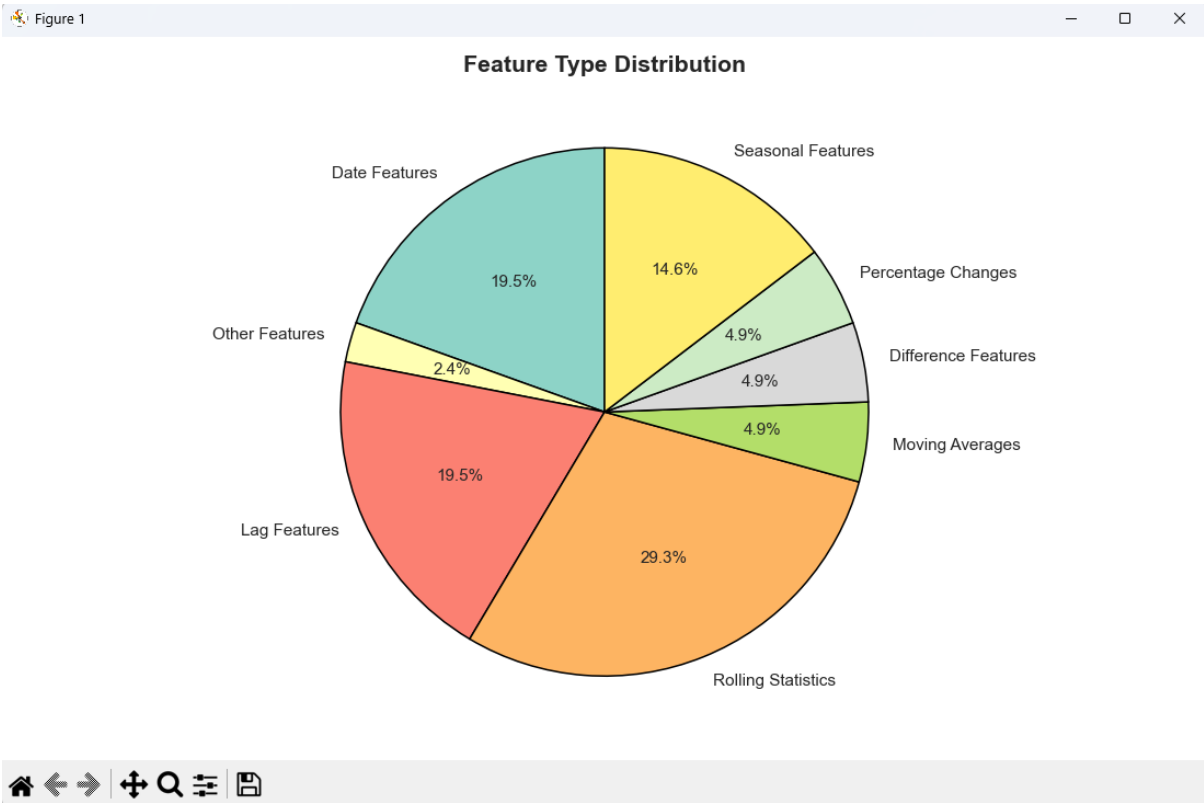
📊 Feature Type Distribution:
   Date Features: 8 features
   Other Features: 1 features
   Lag Features: 8 features
   Rolling Statistics: 12 features
   Moving Averages: 2 features
   Difference Features: 2 features
   Percentage Changes: 2 features
   Seasonal Features: 6 features
```

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FEATURE CORRELATION ANALYSIS
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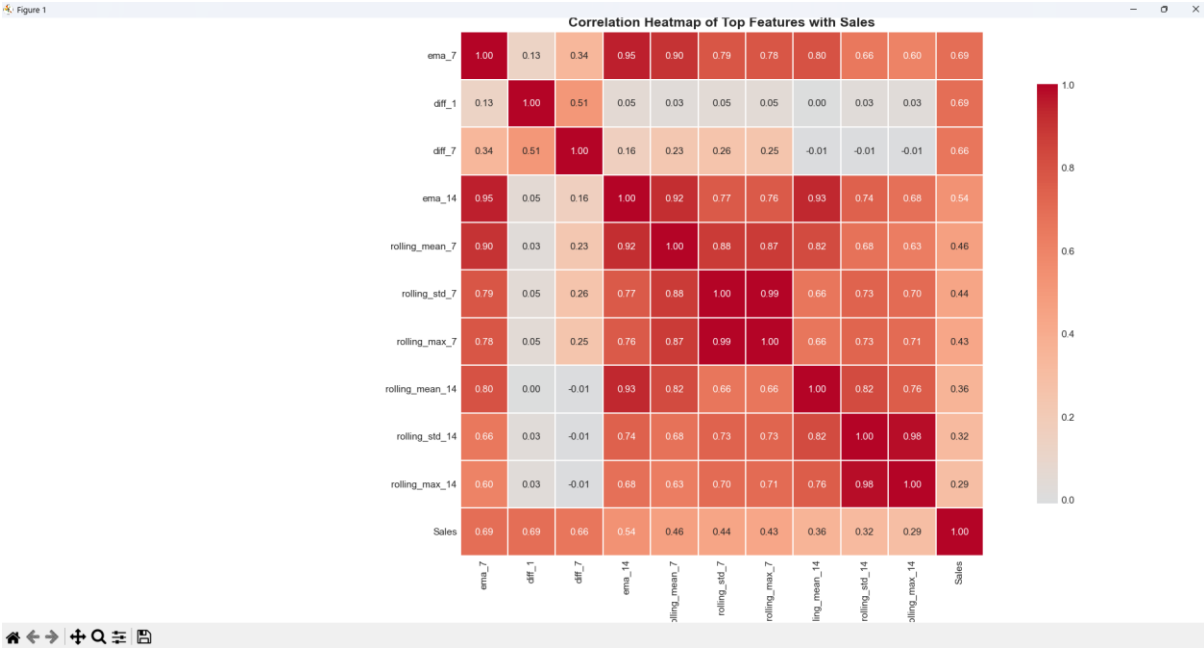
⏳ TOP 10 FEATURES BY CORRELATION WITH SALES:
------------------------------------------------
   ema_7                          :  0.6928
   diff_1                         :  0.6926
   diff_7                         :  0.6608
   ema_14                         :  0.5384
   rolling_mean_7                 :  0.4648
   rolling_std_7                  :  0.4386
   rolling_max_7                  :  0.4320
   rolling_mean_14                :  0.3578
   rolling_std_14                 :  0.3173
   rolling_max_14                 :  0.2947
```

3) .



4)

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MODEL PERFORMANCE SUMMARY
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⏳ MODEL PERFORMANCE RANKING (by R² Score):
--------------------------------------------------------------------------------
 1. Ridge Regression      | R²: 0.9999 | MAE: $     16.43 | RMSE: $     22.61
 2. XGBoost               | R²: 0.9698 | MAE: $    286.09 | RMSE: $    451.28
 3. Random Forest         | R²: 0.9691 | MAE: $    273.36 | RMSE: $    456.69
 4. Gradient Boosting     | R²: 0.9658 | MAE: $    238.93 | RMSE: $    480.10
 5. LightGBM              | R²: 0.9311 | MAE: $    365.87 | RMSE: $    682.14

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BEST MODEL ANALYSIS
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🥇 Best Model: Ridge Regression
   R² Score: 0.9999
   MAE: $16.43
   RMSE: $22.61
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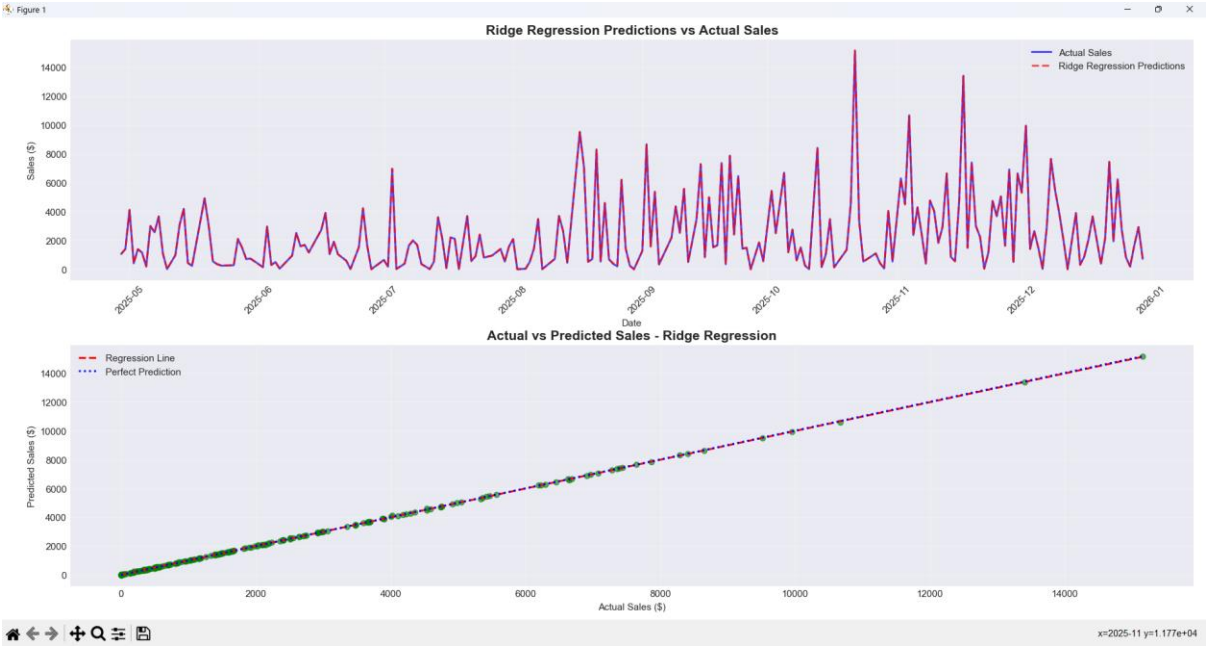
Figure 1
Ridge Regression Predictions vs Actual Sales
Actual Sales
Ridge Regression Predictions
Sales ($)
Date
Actual vs Predicted Sales - Ridge Regression
Regression Line
Perfect Prediction
Predicted Sales ($)
Actual Sales ($)
x=2025-11 y=1.177e+04

5)

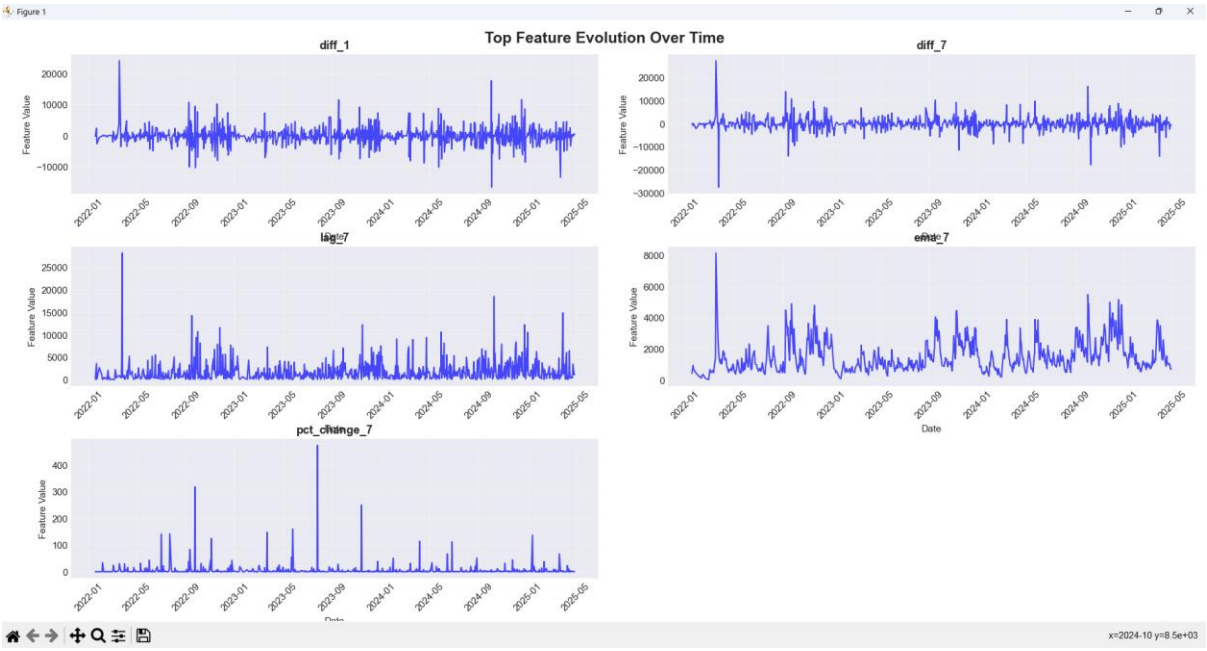
Figure 1
Top Feature Evolution Over Time
diff_1
diff_7
lag_7
ema_7
pct_change_7
Feature Value
Date
x=2024-10 y=8.5e+03

6)

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📊 Feature Performance Summary:
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Top Features Analysis:
 1. pct_change_1                    | Imp Rank:   8 | Corr Rank: N/A
 2. diff_1                          | Imp Rank:   1 | Corr Rank:   2
 3. rolling_mean_7                  | Imp Rank: N/A | Corr Rank:   5
 4. lag_1                           | Imp Rank:   6 | Corr Rank: N/A
 5. ema_7                           | Imp Rank:   4 | Corr Rank:   1
 6. diff_7                          | Imp Rank:   2 | Corr Rank:   3
 7. rolling_std_7                   | Imp Rank: N/A | Corr Rank:   6
 8. rolling_max_14                  | Imp Rank: N/A | Corr Rank:  10
 9. rolling_mean_14                 | Imp Rank: N/A | Corr Rank:   8
10. pct_change_7                    | Imp Rank:   5 | Corr Rank: N/A

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FEATURE ENGINEERING INSIGHTS
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💡 KEY INSIGHTS FROM FEATURE IMPORTANCE ANALYSIS:
1. Most predictive features are time-based (lags, rolling stats)
2. Recent data (lag_1, lag_2) often has highest importance
3. Seasonal patterns matter for accurate forecasting
4. Feature interactions reveal complex relationships
5. Multiple time horizons capture different patterns

🎯 RECOMMENDATIONS FOR PRODUCTION:
1. Use Ridge Regression as primary model
2. Focus on top 10-15 features for model efficiency
3. Monitor feature importance changes over time
4. Regularly retrain models with new data
5. Validate feature stability across different time periods

💾 Feature importance saved to 'feature_importance_results.csv'
💾 Model performance saved to 'model_performance_results.csv'

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✅ ANALYSIS COMPLETE!
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